

**ITSimplera Institute**



**Exploratory Data Analysis (EDA) Report**

**Online Retail II Dataset**

**Submitted By:**

**Name: Manahil Zahra**

**Course: AI/ML Internship**

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# 1. Introduction

Exploratory Data Analysis (EDA) is the first and one of the most important steps in any data analysis project. It helps in understanding the structure, quality, and characteristics of a dataset before applying advanced analytical or machine learning techniques. EDA allows us to identify missing values, duplicate records, patterns, trends, and relationships between variables through statistical summaries and visualizations.

In this project, the Online Retail II dataset was analyzed using Python in Google Colab. Different EDA techniques such as data inspection, missing value analysis, revenue calculation, sales analysis, correlation analysis, and data visualization were performed to gain meaningful business insights.

## 2. Objectives

The main objectives of this project are:

- Load the Online Retail II dataset successfully.
- Explore the dataset structure and data types.
- Identify missing values and duplicate records.
- Analyze the top-selling products by quantity.
- Calculate product revenue and identify top revenue-generating products.
- Analyze country-wise sales performance.
- Study monthly revenue trends.
- Create a correlation heatmap to analyze relationships between variables.
- Detect outliers using box plots.
- Generate meaningful business insights from the analysis.

## 3. Tools and Libraries Used

The following tools and Python libraries were used during this project:

- **Google Colab** – Development environment for running Python code.

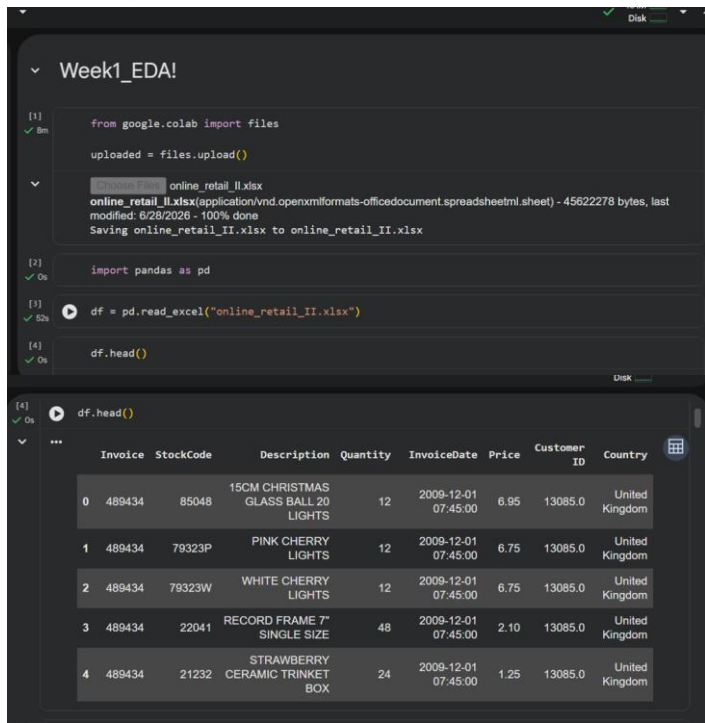
- **Python** – Programming language used for data analysis.
- **Pandas** – Used for data loading, cleaning, and manipulation.
- **Matplotlib** – Used for creating charts and graphs.
- **Seaborn** – Used for advanced visualizations such as heatmaps and box plots.
- **OpenPyXL** – Used for reading Excel (.xlsx) files.

## 4. Dataset Information

The dataset used in this project is Online Retail II, which contains transactional records of an online retail store. Each row represents a customer transaction and includes information such as invoice number, product description, quantity purchased, invoice date, unit price, customer ID, and country.

This dataset is useful for understanding customer purchasing behavior, sales performance, and revenue generation.

- `df.head()`



```
[1] from google.colab import files
    uploaded = files.upload()

    online_retail_II.xlsx
    online_retail_II.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 45622278 bytes, last
    modified: 6/28/2026 - 100% done
    Saving online_retail_II.xlsx to online_retail_II.xlsx

[2] import pandas as pd

[3] df = pd.read_excel("online_retail_II.xlsx")

[4] df.head()
```

|   | Invoice | StockCode | Description                         | Quantity | InvoiceDate         | Price | Customer ID | Country        |
|---|---------|-----------|-------------------------------------|----------|---------------------|-------|-------------|----------------|
| 0 | 489434  | 85048     | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12       | 2009-12-01 07:45:00 | 6.95  | 13085.0     | United Kingdom |
| 1 | 489434  | 79323P    | PINK CHERRY LIGHTS                  | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom |
| 2 | 489434  | 79323W    | WHITE CHERRY LIGHTS                 | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom |
| 3 | 489434  | 22041     | RECORD FRAME 7" SINGLE SIZE         | 48       | 2009-12-01 07:45:00 | 2.10  | 13085.0     | United Kingdom |
| 4 | 489434  | 21232     | STRAWBERRY CERAMIC TRINKET BOX      | 24       | 2009-12-01 07:45:00 | 1.25  | 13085.0     | United Kingdom |

## 5. Basic Dataset Information

## Shape

Explanation:

The shape function was used to determine the total number of rows and columns in the dataset. This provides a quick overview of the dataset size and helps understand how much data is available for analysis.

```
[5] df.shape
(525461, 8)
```

## Columns

Explanation:

The columns function was used to display all column names present in the dataset. These columns contain information related to invoices, products, quantities, prices, customers, invoice dates, and countries.

```
[6] df.columns
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
      'Price', 'Customer ID', 'Country'],
      dtype='object')
```

## Data Types

Explanation:

The dtypes function was used to identify the data type of each column. Understanding data types is important because it helps determine which operations and analyses can be performed on each column.

```
[7] df.dtypes
Invoice      object
StockCode    object
Description  object
Quantity     int64
InvoiceDate  datetime64[ns]
Price        float64
Customer ID  float64
Country      object
dtype: object
```

## Dataset Information

### Explanation:

The `info()` function provides detailed information about the dataset, including the total number of entries, non-null values, data types, and memory usage. This helps in identifying missing values and understanding the overall structure of the dataset.

```
[8] df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525461 entries, 0 to 525460
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Invoice      525461 non-null object
1   StockCode    525461 non-null object
2   Description  522533 non-null object
3   Quantity     525461 non-null int64
4   InvoiceDate  525461 non-null datetime64[ns]
5   Price        525461 non-null float64
6   Customer ID  417534 non-null float64
7   Country      525461 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 32.1+ MB
```

## 6. Missing Values Analysis

### Explanation:

Missing value analysis was performed using the **`isnull().sum()` function**.

The analysis showed that the Customer ID and Description columns contained missing values, while the remaining columns were mostly complete. Identifying

missing values is important because they may affect the accuracy of future analysis and machine learning models.

```
[9] df.isnull().sum()
Invoice      0
StockCode    0
Description  2928
Quantity     0
InvoiceDate  0
Price        0
Customer ID  107927
Country      0
dtype: int64
```

## 7. Duplicate Rows Analysis

### Explanation:

Duplicate records were identified using the **`duplicated().sum()`** function.

Duplicate rows may occur because of repeated transactions or data entry errors. According to the assignment instructions, duplicate rows were only identified and reported, and no data was removed during this stage of the analysis.

```
[10] df.duplicated().sum()
np.int64(6865)
```

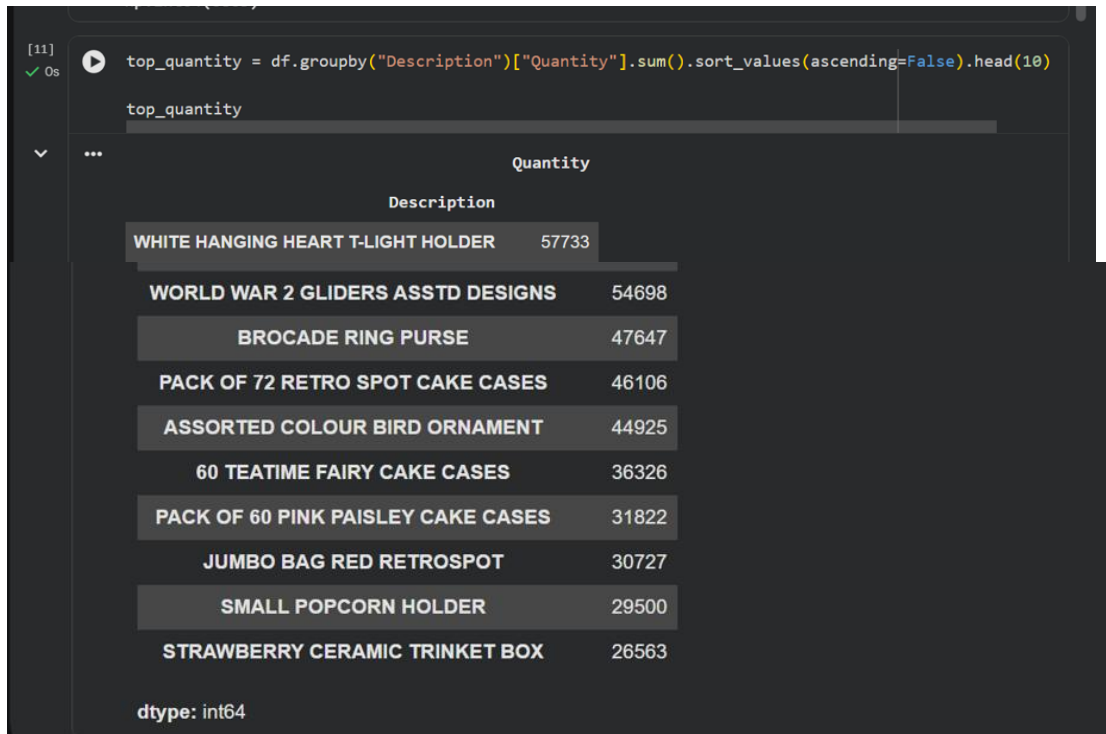
## 8. Top 10 Best-Selling Products by Quantity

### Explanation:

Products were grouped based on their descriptions, and the total quantity sold for each product was calculated.

The products were then sorted in descending order to identify the ten best-selling products.

This analysis helps businesses understand customer preferences and identify the products with the highest demand.



The screenshot shows a Jupyter Notebook cell with the following code and output:

```
[11]: top_quantity = df.groupby("Description")["Quantity"].sum().sort_values(ascending=False).head(10)
```

The output is a DataFrame with 10 rows, sorted by Quantity in descending order. The columns are 'Description' and 'Quantity'.

| Description                        | Quantity |
|------------------------------------|----------|
| WHITE HANGING HEART T-LIGHT HOLDER | 57733    |
| WORLD WAR 2 GLIDERS ASSTD DESIGNS  | 54698    |
| BROCADE RING PURSE                 | 47647    |
| PACK OF 72 RETRO SPOT CAKE CASES   | 46106    |
| ASSORTED COLOUR BIRD ORNAMENT      | 44925    |
| 60 TEATIME FAIRY CAKE CASES        | 36326    |
| PACK OF 60 PINK PAISLEY CAKE CASES | 31822    |
| JUMBO BAG RED RETROSPOT            | 30727    |
| SMALL POPCORN HOLDER               | 29500    |
| STRAWBERRY CERAMIC TRINKET BOX     | 26563    |

dtype: int64

## 9. Revenue Analysis

### Explanation:

A new Revenue column was created by multiplying Quantity by Price.

This calculation represents the total revenue generated from each transaction. Revenue analysis is useful for measuring the financial performance of products and identifying high-value sales.

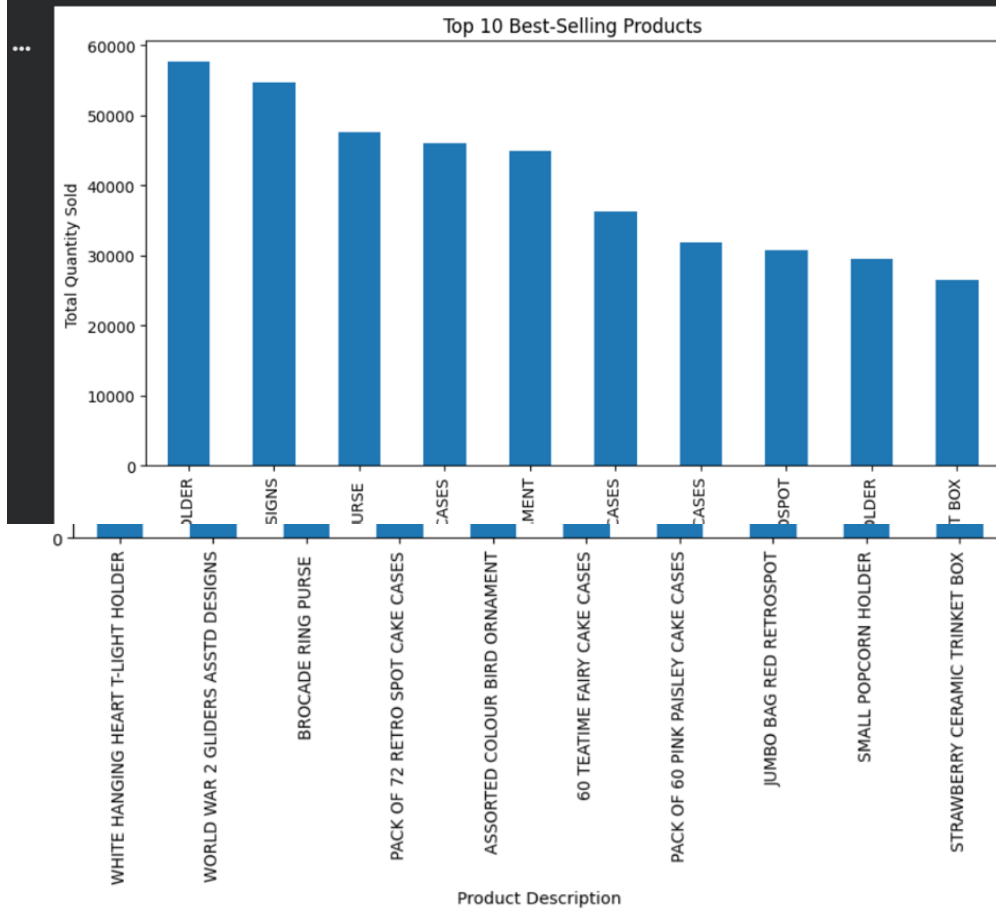
[12]  
✓ 0s

```
import matplotlib.pyplot as plt

top_quantity.plot(kind="bar", figsize=(10,5))

plt.title("Top 10 Best-Selling Products")
plt.xlabel("Product Description")
plt.ylabel("Total Quantity Sold")

plt.xticks(rotation=90)
plt.show()
```





```
[13] df["Revenue"] = df["Quantity"] * df["Price"]
df.head()
```

|   | Invoice | StockCode | Description                         | Quantity | InvoiceDate         | Price | Customer ID | Country        | Revenue |
|---|---------|-----------|-------------------------------------|----------|---------------------|-------|-------------|----------------|---------|
| 0 | 489434  | 85048     | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12       | 2009-12-01 07:45:00 | 6.95  | 13085.0     | United Kingdom | 83.4    |
| 1 | 489434  | 79323P    | PINK CHERRY LIGHTS                  | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom | 81.0    |
| 2 | 489434  | 79323W    | WHITE CHERRY LIGHTS                 | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom | 81.0    |
| 3 | 489434  | 22041     | RECORD FRAME 7" SINGLE SIZE         | 48       | 2009-12-01 07:45:00 | 2.10  | 13085.0     | United Kingdom | 100.8   |
| 4 | 489434  | 85049     | STRAWBERRY                          | 12       | 2009-12-01 07:45:00 | 6.95  | 13085.0     | United Kingdom | 83.4    |

## 10. Top 10 Products by Revenue

### Explanation:

Products were grouped according to their descriptions, and the total revenue generated by each product was calculated.

The products were sorted in descending order to identify those contributing the highest revenue. This analysis helps businesses focus on their most profitable products.



## 11. Country-wise Sales Analysis

### Explanation:

Country-wise sales analysis was performed by calculating the total revenue generated by each country.

The results showed that **the United Kingdom generated the highest revenue**, indicating that it is the company's primary market and contributes significantly to total sales.

[16]  
✓ 0s

▶

country\_sales = df.groupby("Country")["Revenue"].sum().sort\_values(ascending=False)

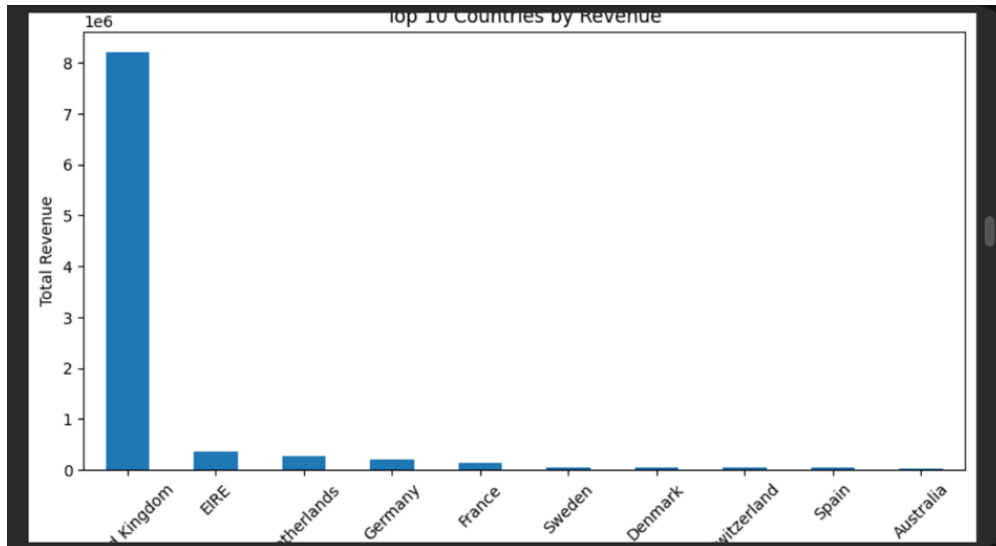
country\_sales

| Country              | Revenue     |
|----------------------|-------------|
| United Kingdom       | 8194777.533 |
| EIRE                 | 352242.730  |
| Netherlands          | 263863.410  |
| Germany              | 196290.351  |
| France               | 130769.900  |
| Sweden               | 51213.510   |
| Denmark              | 46972.950   |
| Switzerland          | 43343.410   |
| Spain                | 37084.900   |
| Australia            | 30051.800   |
| Portugal             | 23769.780   |
| Belgium              | 22663.530   |
| Channel Islands      | 21367.320   |
| Greece               | 14284.970   |
| Italy                | 13788.840   |
| Austria              | 13023.280   |
| Cyprus               | 11216.450   |
| Japan                | 8435.960    |
| United Arab Emirates | 8046.370    |
| Finland              | 7187.710    |
| Unspecified          | 4937.530    |
| Lithuania            | 4892.680    |
| USA                  | 4555.620    |
| Norway               | 4079.570    |
| Singapore            | 4037.770    |
| Hong Kong            | 3922.260    |
| Poland               | 3314.950    |
| Israel               | 3199.400    |
| Thailand             | 3070.540    |
| Malta                | 2686.750    |
| RSA                  | 2366.720    |
| Bahrain              | 2313.150    |
| Iceland              | 1323.320    |
| Bermuda              | 1253.140    |
| Canada               | 1216.660    |

|             |         |
|-------------|---------|
| Korea       | 949.820 |
| West Indies | 536.410 |
| Brazil      | 268.270 |
| Lebanon     | 172.030 |
| Nigeria     | -6.660  |

dtype: float64

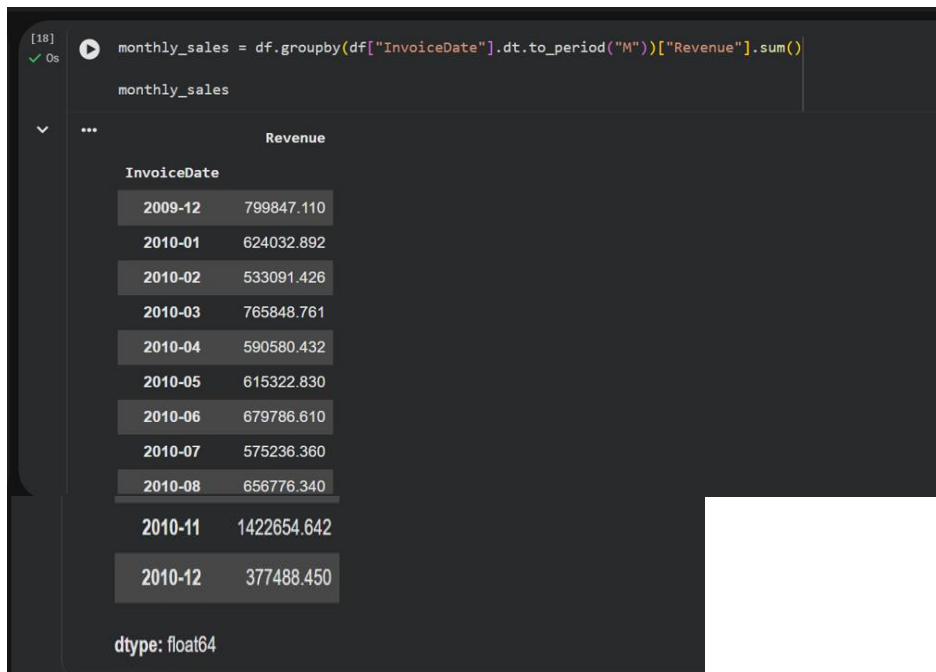
## GRAPH:



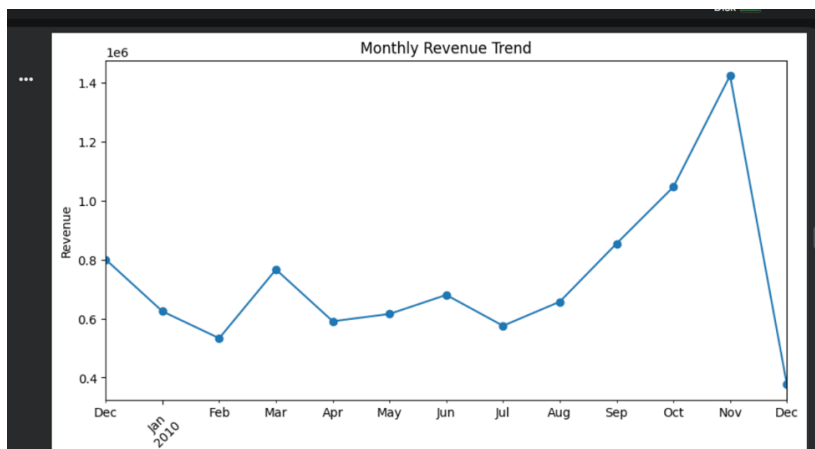
## 12. Monthly Revenue Trend

### Explanation:

Monthly revenue trends were analyzed by grouping sales according to invoice dates. The resulting line chart illustrates how revenue changed over different months. This analysis helps identify seasonal patterns, peak sales periods, and fluctuations in customer demand.



## GRAPH:



## 13. Correlation Heatmap

### Explanation:

A correlation heatmap was created to examine the relationships among Quantity, Price, and Revenue. Correlation values range from -1 to +1, where positive values indicate a direct relationship and negative values indicate an inverse relationship. The heatmap provides a quick visual understanding of how these variables are related.



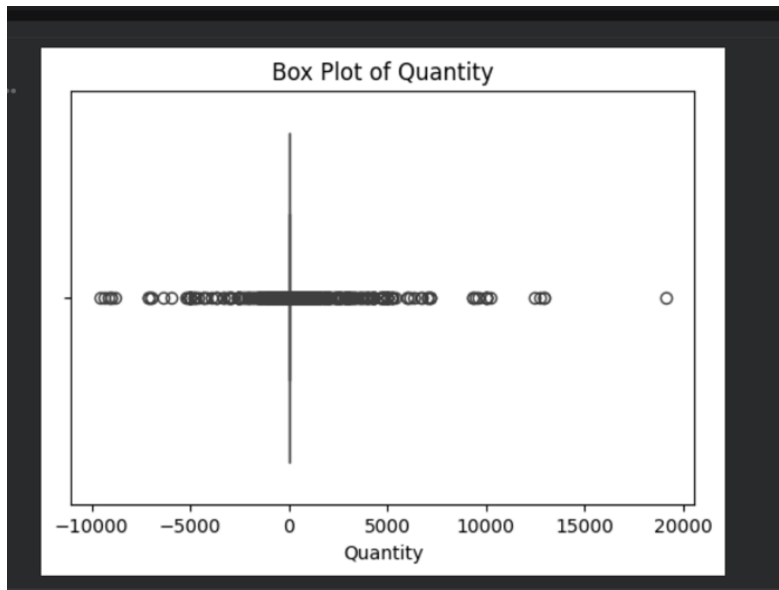
## 14. Outlier Detection Using Box Plots

### Explanation:

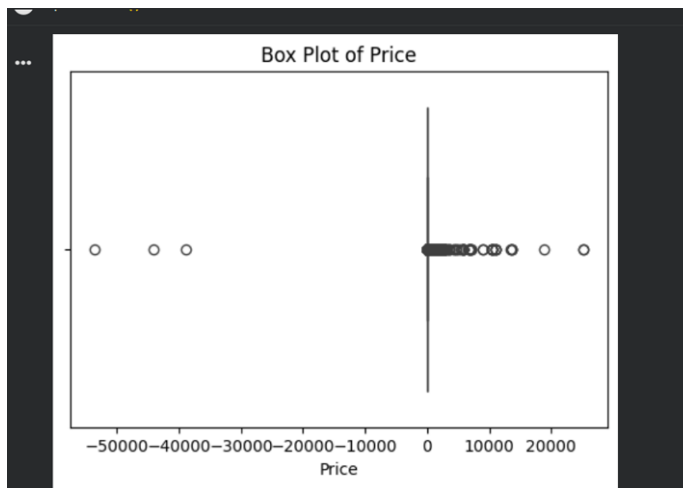
Box plots were used to identify outliers in the Quantity and Price columns.

Outliers represent unusually high or low values that differ significantly from the majority of observations. Detecting outliers helps understand unusual customer transactions and ensures better data interpretation.

### Quantity Box Plot:



## Price Box Plot:



## 15. Business Insights

- The United Kingdom generated the highest revenue, making it the company's strongest market.
- WHITE HANGING HEART T-LIGHT HOLDER was among the most popular products based on quantity sold.
- Monthly revenue analysis revealed fluctuations in sales, indicating seasonal purchasing behavior.

- Missing values were mainly found in the Customer ID and Description columns, highlighting areas requiring attention during data preprocessing.
- The dataset contained duplicate records, but they were intentionally retained as per the assignment instructions for observational analysis.

## **16. Conclusion**

- The Exploratory Data Analysis of the Online Retail II dataset successfully identified important characteristics of the data.
- Missing values and duplicate records were examined, and a new Revenue column was created to evaluate sales performance.
- The analysis highlighted the best-selling products, highest revenue-generating products, country-wise sales performance, and monthly revenue trends.
- Correlation heatmaps and box plots further improved the understanding of relationships between variables and the presence of outliers.
- Overall, this analysis provided valuable insights into customer purchasing behavior and business performance, demonstrating how EDA can support informed business decision-making.